

APPLIED MACHINE LEARNING — COURSE PROJECT

Image-Based Gender Classification: Late Fusion & Knowledge Distillation

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MTCNN Face Detection

ViT Classifier

Late Fusion

Knowledge Distillation

97.2% Accuracy

The Problem & How We Evolved

Starting point

933 TikTok creator videos. No labels. Need a gender classifier.

Strategy: use a pre-trained ViT model as teacher to generate pseudo-labels, then fine-tune a student.

v1 result (naive fine-tuning)

Method	Accuracy
Baseline (majority vote)	91.5%
Fine-tuned (hard labels)	87.9% (-3.6%)

Fine-tuning made things *worse* — standard result when teacher is already near-optimal.

v2 improvements

Method	Accuracy	vs baseline
v2: KD Fine-tuned + LF	97.2%	+5.7%
v1: Baseline (majority vote)	91.5%	—
v1: Hard-label fine-tuned	87.9%	-3.6%

Two changes: better inference (Late Fusion) + better training signal (KD soft labels) = **+5.7%** over baseline.

Pipeline Overview

Two scripts; v2 reuses Step 1 output from v1. All steps checkpointed.

v1: gender_classification.py

- 1 Baseline Inference**
MTCNN + ViT on 4 frames/video -> raw_predictions.csv
- 2 Threshold Sweep + Pseudo-Labels**
6 thresholds scored, best auto-selected, majority vote -> train.csv/test.csv
- 3 Hard-Label Fine-Tuning**
CrossEntropy on 563 videos -> fine_tuned_model/ (87.9%)

v2: gender_classification_v2.py

- A Build KD Splits (thresh=0.90)**
Late fusion soft labels, 666 videos -> train_v2.csv/test_v2.csv
- B KD Fine-Tuning**
KL divergence loss, T=3, soft targets -> kd_model/ (97.2%)
- C Evaluation: All Methods**
Late fusion on all 4 models -> results_v2.txt

Technique 1 — Late Fusion

Why majority vote fails

Example: 4 frames for one video:

Frame	Pred	Confidence
frame_01	female	0.72
frame_02	female	0.71
frame_03	male	0.97
frame_04	male	0.95

Majority vote: 2 female, 2 male — tie! Use mean confidence: female=0.715, male=0.96 — male wins. OK here, but mean confidence tiebreaking doesn't scale well with more frames.

Late fusion: sum probabilities

Frame	P(female)	P(male)
frame_01	0.72	0.28
frame_02	0.71	0.29
frame_03	0.03	0.97
frame_04	0.05	0.95
SUM	1.51	2.49

argmax = male. Correct! High-confidence frames dominate.

No training needed. Just change how you aggregate frame predictions. High-confidence frames automatically dominate.

Technique 2 — Knowledge Distillation

Why hard labels fail

Hard label training treats every sample equally:

Video	Teacher says	Hard label
vid_A	female: 0.99	[1, 0]
vid_B	female: 0.62	[1, 0]

vid_B is likely a noisy label (62% confidence). The model treats it the same as a 99% confident one — it learns wrong things from borderline cases.

Soft labels fix this

Video	Teacher says	Soft label (T=3)
vid_A	female: 0.99	[0.94, 0.06]
vid_B	female: 0.62	[0.55, 0.45]

Temperature T=3 softens the distribution. vid_B now has a [0.55, 0.45] target — the student learns "this is uncertain" instead of treating it as a clean label.

$$L = \text{KL}(\text{student_softmax}/T \parallel \text{teacher_probs}/T) \times T^2$$

Key insight (Hinton et al., 2015): The teacher's probability distribution contains more information than just the label. A "91% female, 9% male" prediction teaches the student something about the structure of the problem.

Confidence Threshold Sweep

Automated threshold selection

Score = balance_score $\times$ $\sqrt{n_videos}$

Rewards both class balance and data quantity.

Thresh	Videos	F	M	Balance	Score
0.70 ✓ v1	704	395	309	0.782	20.76
0.75	697	392	305	0.778	20.54
0.80	687	384	303	0.789	20.68
0.85	677	377	300	0.796	20.71
0.90 (KD)	666	370	296	0.800	20.65
0.95	646	360	286	0.794	20.19

Key observation

Unlike the audio pipeline (where 0.96 removed ALL male samples), the ViT image model is consistently confident across both genders at all thresholds.

All 6 thresholds produce 78-80% class balance. The automated scoring correctly selects 0.70 for v1 (maximum data) and 0.90 is used for KD (cleaner labels for distillation).

Contrast with audio v1: Audio at thresh=0.96 gave 100% female (0 males). Image classifier is far more robust and balanced.

Results — All Methods Compared (n=141 test videos)

87.9%

v1 Hard FT
-3.6% vs baseline

91.5%

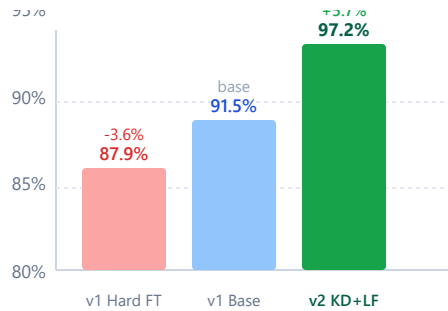
v1 Baseline
reference

97.2%

v2 KD+LF
+5.7% vs baseline

+9.3%

KD gain over
hard-label FT



v1 Baseline confusion matrix

12 errors (female recall 91.1%, male recall 91.9%)

	Pred F	Pred M
True F (79)	72	7
True M (62)	5	57

v2 KD fine-tuned confusion matrix +5.7%

4 errors (female recall 96.2%, male recall 98.4%)

	Pred F	Pred M
True F (79)	76	3
True M (62)	1	61

Why KD + Late Fusion Gains +5.7%?

Why Late Fusion helps

The ViT baseline is already **98.7% accurate on clean faces**. Errors in v1 (91.5%) come from:

- Low-quality frames (motion blur, side profile)
- Majority vote giving equal weight to weak and strong predictions
- Tie-breaking with mean confidence is fragile

Late fusion fixes this: a 0.99 frame contributes $P(\text{female})=0.99$, a 0.71 frame contributes 0.71 — high-confidence frames dominate naturally.

Why KD helps over hard-label FT

- Hard labels treat every sample as equally certain — noisy pseudo-labels corrupt the model
- Soft targets ($T=3$) encode uncertainty — a $[0.55, 0.45]$ target says "this is borderline", not "definitely female"
- Stricter threshold (0.90) filters out the noisiest pseudo-labels before distillation

Result: errors drop from 12 (v1 baseline) to 4 (KD+LF) — a **+5.7% accuracy gain** and **+9.3% over hard-label fine-tuning**.

Conclusion

Summary of contributions

- 1 MTCNN face detection on pre-extracted TikTok frames
- 2 Automated threshold sweep (balance $\times \sqrt{n}$ scoring)
- 3 **Late Fusion:** sum softmax probs across frames — high-confidence frames dominate, eliminates tie-breaking fragility
- 4 **Knowledge Distillation:** soft labels + KL loss + $T=3 \rightarrow 97.2\%$ (+5.7% vs baseline, +9.3% vs hard FT)
- 5 Image-based (97.2%) vastly outperforms audio-based (86.3%)

All edge cases handled

- Video < 4 frames — use however many exist
- Frame has no face — skip frame silently
- All frames no face — log to skipped_videos.txt
- Tie vote $\rightarrow$ resolved by mean confidence
- < 10 class samples $\rightarrow$ skip fine-tuning with warning
- No face at training time $\rightarrow$ black image fallback

Bottom line: Combining late fusion at inference with KD soft-label training yields a **+5.7% accuracy gain** over the baseline, going from 91.5% to 97.2% on 141 held-out test videos.